PAWS

Automatic Feeder

What is PAWS?

PAWS is an automated cat feeder I designed, which I plan to replicate across campus with the help of my juniors in Animo SPEAR. Our campus is home to over 10 cats, which is why food dispensers are placed in almost every corner. However, prolonged exposure to the elements often leaves these open feeders dirty and infested with flies.

Moving forward, I want to upgrade this prototype into a larger, more durable version—either 3D-printed or constructed from stainless steel to prevent rust. Future plans also include integrating an onboard camera and adding extra smart features.

Why?

I truly love cats! In fact, I am a member of NEKO, our school's cat welfare organization, where I serve on both the Cat-Caretaking and Activities & Management committees. Every week, I am assigned shifts to refill food bowls across campus. It was during these shifts that I noticed the exact flaws and inefficiencies I described earlier.

What pushed me to pursue this?



Figure 1. Situation of the current feeders

During my NEKO shifts I saw open bowls left dirty and full of flies. Our campus has over 10 cats, so I built a closed feeder.

The main body that holds the food is repurposed from a free milk-brand food container my mom no longer uses. The rest of the structure is cardboard painted black, and a servo flap controls the opening.



Figure 2. Creating the dispensing opening, which will be covered by a flap controlled by a servo motor.

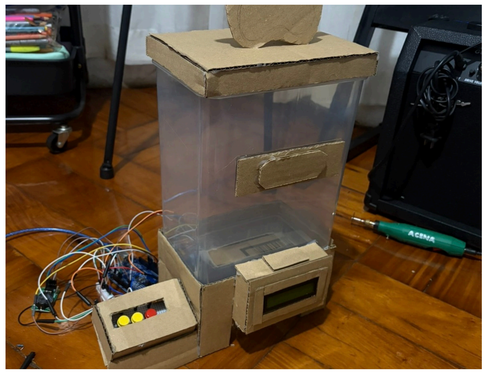


Figure 3. The earlier prototype

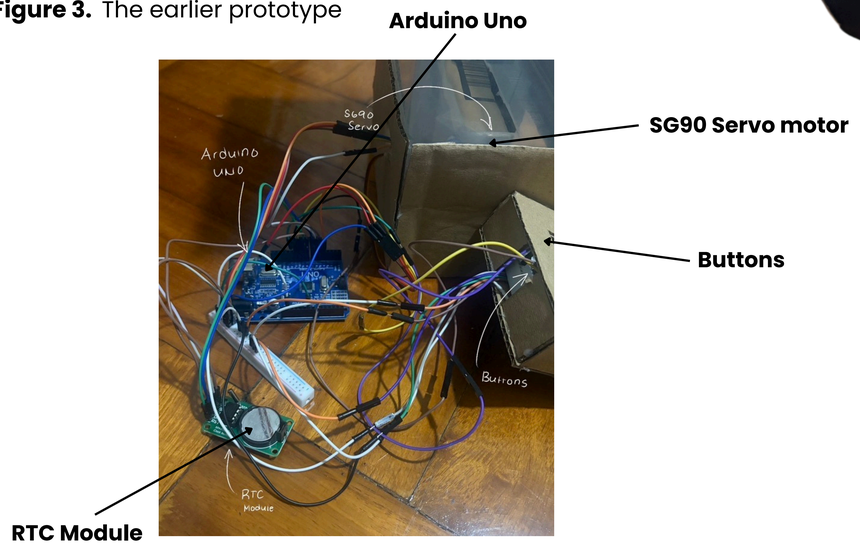


Figure 4. What's inside the PAWS

I absolutely love arcades! Whenever I hang out with friends, visiting places like Timezone or Tom's World is always on our itinerary. Because of this passion, I designed the feeder to look like a claw machine!



The parts

SG90 servo, Arduino Uno, Buttons, and 9V battery

Status

Working prototype. It feeds automatically every 3 hours and gives about 50 g of dry food each time. Three buttons let you add hours, add minutes, or feed manually, so the schedule can change for cats that should eat less. It has already fed the cats on campus. Next: a larger 3D-printed or stainless steel version, and a camera.